

Exercise 9C Solutions

Percentage Change & Its Applications

Key Concept: *Percentage change (increase or decrease) is always calculated on the **Original Value**.*

$$\text{Percentage Change} = \frac{\text{Change in Value}}{\text{Original Value}} \times 100\%$$

- To increase a number x by $p\%$, we find $x \times \left(1 + \frac{p}{100}\right)$. - To decrease a number x by $p\%$, we find $x \times \left(1 - \frac{p}{100}\right)$.

Question 1: Increase

Increase:

- (i) 375 by 4%
- (i) 500 by 3.4%
- (i) 70 by 140%
- (i) 48 by $12\frac{1}{2}\%$
- (i) 90 by 2%

Solutions for Question 1

(i) **Increase 375 by 4%:**

$$\text{Increase} = 4\% \text{ of } 375 = \frac{4}{100} \times 375 = \frac{1500}{100} = 15$$

$$\text{Increased Number} = 375 + 15 = 390$$

(i) **Increase 500 by 3.4%:**

$$\text{Increase} = 3.4\% \text{ of } 500 = \frac{3.4}{100} \times 500 = 3.4 \times 5 = 17$$

$$\text{Increased Number} = 500 + 17 = 517$$

(i) **Increase 70 by 140%:**

$$\text{Increase} = 140\% \text{ of } 70 = \frac{140}{100} \times 70 = 14 \times 7 = 98$$

$$\text{Increased Number} = 70 + 98 = 168$$

(i) **Increase 48 by $12\frac{1}{2}\%$:**

$$\text{Percentage} = 12\frac{1}{2}\% = \frac{25}{2}\%$$

$$\text{Increase} = \frac{25}{200} \times 48 = \frac{1}{8} \times 48 = 6$$

$$\text{Increased Number} = 48 + 6 = 54$$

(i) **Increase 90 by 2%:**

$$\text{Increase} = 2\% \text{ of } 90 = \frac{2}{100} \times 90 = 1.8$$

$$\text{Increased Number} = 90 + 1.8 = 91.8$$

Question 2: Decrease

Decrease:

- (i) 70 by 40%
- (i) 340 by 35%
- (i) 65 by 4%
- (i) 36 by $16\frac{2}{3}\%$
- (i) 260 by 1.5%

Solutions for Question 2**(i) Decrease 70 by 40%:**

$$\text{Decrease} = 40\% \text{ of } 70 = \frac{40}{100} \times 70 = 28$$

$$\text{Decreased Number} = 70 - 28 = 42$$

(i) Decrease 340 by 35%:

$$\text{Decrease} = 35\% \text{ of } 340 = \frac{35}{100} \times 340 = \frac{11900}{100} = 119$$

$$\text{Decreased Number} = 340 - 119 = 221$$

(i) Decrease 65 by 4%:

$$\text{Decrease} = 4\% \text{ of } 65 = \frac{4}{100} \times 65 = 2.6$$

$$\text{Decreased Number} = 65 - 2.6 = 62.4$$

(i) Decrease 36 by $16\frac{2}{3}\%$:

$$\text{Percentage} = 16\frac{2}{3}\% = \frac{50}{3}\%$$

$$\text{Decrease} = \frac{50}{300} \times 36 = \frac{1}{6} \times 36 = 6$$

$$\text{Decreased Number} = 36 - 6 = 30$$

(i) Decrease 260 by 1.5%:

$$\text{Decrease} = 1.5\% \text{ of } 260 = \frac{1.5}{100} \times 260 = 1.5 \times 2.6 = 3.9$$

$$\text{Decreased Number} = 260 - 3.9 = 256.1$$

Question 3: Find a Number (Increase Case)

Find a number:

- (i) Which when increased by 10% becomes 66.
- (i) Which when increased by 120% becomes 77.
- (i) Which when increased by 2.5% becomes 246.

Solutions for Question 3

Let the required number be x .

(i) **Increased by 10% becomes 66:**

$$x + 10\% \text{ of } x = 66 \implies x + 0.1x = 66 \implies 1.1x = 66$$

$$x = \frac{66}{1.1} = 60$$

The required number is **60**.

(i) **Increased by 120% becomes 77:**

$$x + 120\% \text{ of } x = 77 \implies x + 1.2x = 77 \implies 2.2x = 77$$

$$x = \frac{77}{2.2} = \frac{770}{22} = 35$$

The required number is **35**.

(i) **Increased by 2.5% becomes 246:**

$$x + 2.5\% \text{ of } x = 246 \implies x + 0.025x = 246 \implies 1.025x = 246$$

$$x = \frac{246}{1.025} = 240$$

The required number is **240**.

Question 4: Find a Number (Decrease Case)

Find a number:

- (i) Which when decreased by 35% becomes 52.
- (i) Which when decreased by 8% becomes 115.
- (i) Which when decreased by 3.4% becomes 483.

Solutions for Question 4Let the required number be x .

- (i) Decreased by 35% becomes 52:**

$$x - 35\% \text{ of } x = 52 \implies 0.65x = 52$$

$$x = \frac{52}{0.65} = \frac{5200}{65} = 80$$

The required number is **80**.

- (i) Decreased by 8% becomes 115:**

$$x - 8\% \text{ of } x = 115 \implies 0.92x = 115$$

$$x = \frac{115}{0.92} = \frac{11500}{92} = 125$$

The required number is **125**.

- (i) Decreased by 3.4% becomes 483:**

$$x - 3.4\% \text{ of } x = 483 \implies (1 - 0.034)x = 483 \implies 0.966x = 483$$

$$x = \frac{483}{0.966} = \frac{483000}{966} = 500$$

The required number is **500**.**Question 5 & 6: Multiplier Rule**

- 5. By what number must a given number be multiplied to increase it by 12%?
- 6. By what number must a given number be multiplied to decrease it by 30%?

Solutions for Questions 5 & 6

5. Let the given number be
- x
- .

$$\text{Increased value} = x + 12\% \text{ of } x = x + 0.12x = 1.12x$$

Thus, the given number must be multiplied by **1.12** to increase it by 12%.

6. Let the given number be
- x
- .

$$\text{Decreased value} = x - 30\% \text{ of } x = x - 0.30x = 0.70x$$

Thus, the given number must be multiplied by **0.7** (or **0.70**) to decrease it by 30%.

Question 7: Price Percentage Increase

The price of a fan increases from Rs 3260 to Rs 3749. Find the increase per cent in its price.

Solution for Question 7

$$\text{Original Price} = \text{Rs } 3260$$

$$\text{Increased Price} = \text{Rs } 3749$$

$$\text{Increase in Price} = 3749 - 3260 = \text{Rs } 489$$

$$\text{Increase Per Cent} = \frac{\text{Increase}}{\text{Original Price}} \times 100\% = \frac{489}{3260} \times 100\%$$

$$\text{Increase Per Cent} = \frac{4890}{326}\% = 15\%$$

The price of the fan increased by **15%**.

Question 8: Provident Fund Deduction

The monthly salary of Mr Rakesh is Rs 32500. After deducting the provident fund, he gets Rs 29900 per month. What per cent of the salary is deducted as provident fund?

Solution for Question 8

$$\text{Total Monthly Salary} = \text{Rs } 32500$$

$$\text{Salary Received After Deduction} = \text{Rs } 29900$$

$$\text{Amount Deducted} = 32500 - 29900 = \text{Rs } 2600$$

$$\text{Deduction Per Cent} = \frac{\text{Amount Deducted}}{\text{Total Salary}} \times 100\%$$

$$\text{Deduction Per Cent} = \frac{2600}{32500} \times 100\% = \frac{2600}{325}\% = 8\%$$

The percentage of salary deducted as provident fund is **8%**.

Question 9: Car Depreciation Rate

A car was purchased last year for Rs 415000. Now, its value is Rs 356900. At what rate is the car depreciating?

Solution for Question 9

$$\text{Original Value (Last Year)} = \text{Rs } 415000$$

$$\text{Current Value} = \text{Rs } 356900$$

$$\text{Depreciation (Decrease in Value)} = 415000 - 356900 = \text{Rs } 58100$$

$$\text{Depreciation Rate (Per Cent)} = \frac{\text{Depreciation}}{\text{Original Value}} \times 100\%$$

$$\text{Depreciation Rate} = \frac{58100}{415000} \times 100\% = \frac{581}{41.5}\% = \frac{5810}{415}\% = 14\%$$

The car is depreciating at a rate of **14% per annum**.

Question 10: Finding Original Price of Car

On decreasing the price of a car by 6%, its value becomes Rs 249100. What was the original price of the car?

Solution for Question 10

Let the original price of the car be Rs x .

$$\text{Decreased Price} = (100\% - 6\%) \text{ of } x = 94\% \text{ of } x$$

$$94\% \text{ of } x = 249100 \implies \frac{94}{100} \times x = 249100$$

$$x = 249100 \times \frac{100}{94} = 2650 \times 100 = 265000$$

The original price of the car was **Rs 2,65,000**.

Question 11: Finding Original Salary

On increasing the salary of a man by 12%, his salary is increased by Rs 2316. What was his original salary?

Solution for Question 11

Let the original salary be Rs x .

The increase in salary is 12% of the original salary.

$$12\% \text{ of } x = 2316 \implies \frac{12}{100} \times x = 2316$$

$$x = 2316 \times \frac{100}{12} = 193 \times 100 = 19300$$

The original salary of the man was **Rs 19,300**.

Question 12: Successive Percentage Change (Net Salary Change)

The salary of Gopal was increased by 10% and then the increased salary was decreased by 10%. Find the net increase or decrease per cent in his original salary.

Solution for Question 12

Let Gopal's original salary be Rs 100.

Step 1: Apply the 10% increase:

$$\text{Increased Salary} = 100 + 10\% \text{ of } 100 = 100 + 10 = \text{Rs } 110$$

Step 2: Apply the 10% decrease to the increased salary:

$$\text{Decreased Salary} = 110 - 10\% \text{ of } 110 = 110 - 11 = \text{Rs } 99$$

Now, compare the final salary with the original salary:

$$\text{Net Decrease in Value} = \text{Original Salary} - \text{Final Salary} = 100 - 99 = \text{Rs } 1$$

$$\text{Net Decrease Per Cent} = \frac{\text{Net Decrease}}{\text{Original Salary}} \times 100\% = \frac{1}{100} \times 100\% = 1\%$$

Therefore, there is a **net decrease of 1%** in his original salary.

Question 13: Bill Deduction

After deducting 4% of a bill, the amount still to be paid is Rs 1488. How much was the original bill?

Solution for Question 13

Let the original bill amount be Rs x .

$$\text{Remaining amount to be paid} = (100\% - 4\%) \text{ of } x = 96\% \text{ of } x$$

$$96\% \text{ of } x = 1488 \implies \frac{96}{100} \times x = 1488$$

$$x = 1488 \times \frac{100}{96} = 15.5 \times 100 = 1550$$

The original bill amount was **Rs 1,550**.

Question 14: Error Per Cent in Measurement

The weight of a boy was 40 kg. But, it was wrongly measured as 42 kg. Find the error per cent.

Solution for Question 14

$$\text{Actual Weight} = 40 \text{ kg}$$

$$\text{Measured Weight} = 42 \text{ kg}$$

$$\text{Error in Measurement} = 42 - 40 = 2 \text{ kg}$$

$$\text{Error Per Cent} = \frac{\text{Error}}{\text{Actual Weight}} \times 100\%$$

$$\text{Error Per Cent} = \frac{2}{40} \times 100\% = \frac{1}{20} \times 100\% = 5\%$$

The measurement error is **5%**.